Sri Chaitanya IIT Academy.,India.
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A right Choice for the Real Aspirant
ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN_Shift-1_Batch-II | Date: 20-01-2026
Time: 09.00Am to 12.00Pm | GTM-16 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**3**	3	**3**	4	**2**	5	**1**
6	**1**	7	**1**	8	**1**	9	**3**	10	**3**
11	**4**	12	**1**	13	**2**	14	**4**	15	**2**
16	**3**	17	**1**	18	**3**	19	**1**	20	**2**
21	**2**	22	**202**	23	**19**	24	**5**	25	**40**

PHYSICS

26	**3**	27	**4**	28	**1**	29	**3**	30	**3**
31	**1**	32	**4**	33	**2**	34	**3**	35	**3**
36	**3**	37	**2**	38	**1**	39	**1**	40	**3**
41	**1**	42	**2**	43	**1**	44	**2**	45	**1**
46	**2**	47	**106**	48	**3**	49	**9**	50	**4**

CHEMISTRY

51	**4**	52	**4**	53	**1**	54	**2**	55	**2**
56	**2**	57	**3**	58	**4**	59	**4**	60	**3**
61	**3**	62	**2**	63	**1**	64	**1**	65	**2**
66	**1**	67	**2**	68	**4**	69	**3**	70	**3**
71	**86**	72	**25**	73	**15**	74	**100**	75	**3**

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. **Clearly ,** $B(x:x\in Z \text{ and } x^2 \text{ is even})$

$\Rightarrow B(1,4,9,16,.....)$ Which is an infinite set.

2.

i) Let $a\in N$

$\because a \text{ is a factor of } a\Rightarrow (a,a)\in R\forall a\in N$

$\Rightarrow R \text{ is reflecxive}$

$ii)\text{let } a,b,c\in N \text{ and } (a,b)\in R,(b,c)\in R$

$(a,b)\in R,(b,c)\in R$

$\Rightarrow a \text{ is a factor } b \text{ and } b \text{ is a factor of } c$

$\Rightarrow b=ad, c=be \text{ when } d,e\in N$

$\Rightarrow c=(ad)e=a(de)$

$\Rightarrow a \text{ is a factor of } c$

$\Rightarrow (a,c)\in R$

$iii)\text{Above relation } R \text{ is not symmetric; for example,}$

$2,6\in N \text{ and } 2 \text{ is a factor of } 6.$

$\Rightarrow (2,6)\in R$

$\text{but } 6 \text{ is not a factor of } 2.$

$\Rightarrow (6,2)\in R$

$\text{Hence, } R \text{ is not symmetric}$

3. .

We know that

$$\sin\theta.\sin\left(60^\circ-\theta\right).\sin\left(60^\circ+\theta\right)=\frac{1}{4}\sin 3\theta$$

Therefore,

$$\sin 20^\circ.\sin 40^\circ.\sin 80^\circ=\frac{1}{4}\sin 60^\circ=\frac{\sqrt{3}}{8} \text{ and}$$

$$\sin 20^\circ.\sin 40^\circ.\sin 60^\circ.\sin 80^\circ=\frac{\sqrt{3}}{8}\sin 60^\circ=\frac{3}{16}$$

4. .

$$\sin\left(\frac{\pi}{4}\cot\theta\right)=\sin\left(\frac{\pi}{2}-\frac{\pi}{4}\tan\theta\right)$$

$$\frac{\pi}{4}(\cot\theta+\tan\theta)=\frac{\pi}{2}$$

$$\cot\theta+\tan\theta=2 \quad \text{or} \quad \sin 2\theta=1$$

$$\Rightarrow 2\theta=2n\pi+\frac{\pi}{2} \qquad \text{(or)}$$

$$\theta=n\pi+\frac{\pi}{4}, n\in \mathbf{Z}$$

5. .

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$$1+\cos 2\alpha + i\sin 2\alpha$$

$$= 2\cos^2\alpha + 2i\sin\alpha\cos\alpha$$

$$= 2\cos\alpha(\cos\alpha + i\sin\alpha)$$

$$= -2\cos\alpha\left((\cos(\pi+\alpha) + i\sin(\pi+\alpha)\right)$$

observe that $\cos\alpha < 0$.

6. .

$$\left(x^2+x-2\right)\left(x^2+x-3\right)=12$$

$$\Rightarrow (y-2)(y-3)=12 \text{ where } y = x^2 + x$$

$$\Rightarrow y^2-5y-6=0 \Rightarrow y=6,\ -1$$

$$\Rightarrow x^2+x=6,\ x^2+x=-1$$

$x^2+x=6 \Rightarrow x=-3,2$ *are real roots*

and $x^2+x+1=0$ *has complex roots*

$$\frac{-1\pm\sqrt{3}i}{2}$$

$\therefore$ *sum of non – real roots*

$$=\left(\frac{-1+\sqrt{3}i}{2}\right)+\left(\frac{-1-\sqrt{3}i}{2}\right)=-1$$

7. .

The number of way of selection of pairs of balls consists one black ball and one white ball

$$=\left({}^{n}C_1\right)\left({}^{n}C_1\right)\left({}^{n-1}C_1\right)\left({}^{n-1}C_1\right).....\left({}^{1}C_1\right)\left({}^{1}C_1\right)$$

$$=\left({}^{n}C_1\ {}^{n-1}C_1\ {}^{n-2}C_1.....{}^{1}C_1\right)^2 = 576$$

$[n(n-1)(n-2)\cdots 1] = (24) \Rightarrow n! = 24 = 4!$

$\therefore\ n = 4.$ **A1**

8. 1) it is given that the 10th place of 5 digit number with distinct digits is '2', then the ten thousand place, thousand place, hundred place and unit place we can fill in 8,8,7 and 6 ways only.

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So, required number of five diti number is

8 x 8 x 7 x 6 = 336 k(given)

$\Rightarrow k = \frac{8\times8\times7\times6}{336} = \frac{8\times8\times7}{56} = 8$

9.

The sum of the coefficients in the expansion

$g(x) = \left(k^2x^2 - 6kx + 5\right)^{37}$ is $\left(g(1)\right)^n = 0$

$\left(k^2 - 6k + 5\right)^{37} = 0 \Rightarrow k^2 - 6k + 5 = 0$

$= (k-1)(k-5) = 0$

$\therefore\ k = 1$ or 5

10. .sum of infinite G.P is $\frac{a}{1-r}$.

$x = \frac{1}{1-a} \Rightarrow a = 1 - \frac{1}{x},$

$y = \frac{1}{1-b} \Rightarrow b = 1 - \frac{1}{y},$

$z = \frac{1}{1-c} \Rightarrow c = 1 - \frac{1}{z},$

a,b,c *are in A.P* $\Rightarrow 1-\frac{1}{x}, 1-\frac{1}{y}, 1-\frac{1}{z}$ *are in A.P*

$\Rightarrow -\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z}$ *are in A.P*

$\Rightarrow x, y, z$ *are in H.P.*

11. .

(1, 2), (3, 4) to lie on the same side

$\Rightarrow \frac{L_{11}}{L_{22}} > 0$

i.e., $\frac{3-10+a}{9-20+a} > 0$

$\Rightarrow \frac{a-7}{a-11} > 0 \Rightarrow (a-7)(a-11) > 0$

$\Rightarrow a < 7$ or $a > 11$

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12. .

. Given equation represents a circle if

$g^2 + 16 - (4g^2 - 7g + 18) \geq 0$

$g^2 + 16 - 4g^2 + 7g - 18 \geq 0$

$-3g^2 + 7g - 2 \geq 0$ ...(1)

The roots of $3g^2 - 7g + 2 = 0$ are $1/3, 2$.

The sign of quadratic expression $-3g^2 + 7g - 2$ and the coefficient of x^2 are opposite in signs.

$\therefore g$ should lie between $\left[\frac{1}{3}, 2\right]$

13. .

Equation can be written as

$(y - 6)^2 + 8x - 16 = 0$

$(y - 6)^2 = -8(x - 2)$

$(y - 6)^2 = -4(2)(x - 2)$

Focus $= (-a + h, k) = (-2 + 2, 6) = (0, 6)$

14. .

Sol: Given ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1 \Rightarrow a^2 = 4,\ b^2 = 3$

$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow e = \frac{1}{2}$

$\therefore$ Focus S $= (ae, 0) = (1, 0)$

Let P$(\theta) = (a\cos\theta, b\sin\theta)$

$= \left(2\cos\theta, \sqrt{3}\sin\theta\right)$

given that $SP = \frac{5}{4}$

$(2\cos\theta - 1)^2 + 3\sin^2\theta = \frac{25}{16}$

$(2 - \cos\theta)^2 = \frac{25}{16}$

$2 - \cos\theta = \frac{5}{4} \Rightarrow \cos\theta = \frac{3}{4}$

$\Rightarrow \theta = \cos^{-1}\frac{3}{4}$ or $2\pi - \cos^{-1}\left(\frac{3}{4}\right)$

15. .

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put $\text{Cos}^{-1}x = t$ then $\text{cost} = x$

$t \to \pi$ as $x \to -1$

(observe that the function is defined only in a right neighbourhood of -1).

$$\lim_{t\to\pi^-} \frac{\sqrt{\pi}-\sqrt{t}}{\sqrt{1+\cos t}} = \lim_{t\to\pi^-}\left[\frac{1}{\sqrt{\pi}+\sqrt{t}}\frac{\pi-t}{\sqrt{2}\cos\frac{t}{2}}\right]$$

$$= \frac{2}{2\sqrt{2}\sqrt{\pi}} \lim_{\frac{\pi}{2}-t\to 0} \frac{\frac{\pi}{2}-\frac{t}{2}}{\sin\left(\frac{\pi}{2}-\frac{t}{2}\right)}$$

$$= \frac{1}{\sqrt{2\pi}}$$ **Ans: (2)**

16. For n consequence integers, the mean deviation about the mean

$$= \frac{n^2-1}{4n} = \frac{8(n+1)}{n} \Rightarrow n = 33$$

17. a) the projection of line joining the points P (7, -5, 11) and Q(-2,8,13) on a line with direction cosines $\frac{1}{3},\frac{2}{3},\frac{2}{3}$ is

$$= (-2-7)\frac{1}{3}+(8+5)\frac{2}{3}+(13-11)\frac{2}{3}$$

$$= -3+\frac{26}{3}+\frac{4}{3} = 7$$

18. .Conceptual

19. .

$$(a)\quad \tan^{-1}\frac{1}{1+n+n^2} = \tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right)$$

$$= \tan^{-1}(n+1) - \tan^{-1} n$$

$$\text{So, } \sum_{n=1}^{50}(\tan^{-1}(n+1) - \tan^{-1} n)$$

$$= \tan^{-1} 51 - \tan^{-1} 1$$

$$\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right) = \cot(\tan^{-1}51 + \tan^{-1}1)$$

$$= \frac{1}{\tan(\tan^{-1}51 - \tan^{-1}1)} = \frac{1+51\times1}{51-1} = \frac{52}{50} = \frac{26}{25}$$

20. . Conceptual
21. Conceptual
22. Conceptual
23. Conceptual
24. Conceptual
25. Conceptual

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PHYSICS

26. .

$T = m_1 g ---(1)$

$T = m_2 g + F_e ------(2)$

$from\ (1)\ \&\ (2)\ m_1 g = m_2 g + F_e \Rightarrow m_1 g = m_2 g + \frac{q^2}{\epsilon_0 A} \Rightarrow m_1 = m_2 + \frac{q^2}{2\epsilon_0 Ag}$

27. .

$7\Omega\ is\ short\ circuit\ R_{eq} = 4 \times \frac{1}{4} = 1\Omega$

28. CONCEPTUAL

29. .

$z = \frac{E}{I} = \frac{12}{0.5} = 24\Omega\ \ \&\ \ z = \sqrt{R^2 + w^2 L^2}\ \ \&\ \ w = 2\pi n$

$then\ find\ L = \sqrt{\frac{V^2 - R^2}{R^2}} = \frac{12\sqrt{3}}{314} = 0.066H$

30. . .

After P length of pendulum $\frac{1}{4} \Rightarrow T^1 = \frac{T}{2}; t = \frac{T}{2} + \frac{T^1}{2} = \frac{T}{2} + \frac{T}{4} = \frac{3T}{4}$

31. CONCEPTUAL

32. CONCEPTUAL

33. .

$\Delta y = (y_{11})_{dark} - (y_7)_{bright} = \frac{(2 \times 11 - 1) \times D}{2d} = 8.75 \times 10^{-3} m = 8.75mm$

34. ***To evaluate the assertion and reason provided, we need to consider the formula for the density of a nucleus. The density ρ can be expressed as follows:***

$$\rho = \frac{M}{V} = \frac{m_n \times A}{\frac{4}{3}\pi R^3}$$

Here, m_n represents the nucleon mass, A is the mass number, and R is the radius of the nucleus. According to the formula for the radius:

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$$R = R_0 A^{1/3}$$

This indicates that the radius R is proportional to $A^{1/3}$. By substituting the expression for R in terms of A back into the formula for density, we get:

$$\rho = \frac{m_n}{\frac{4}{3}\pi R_0^3}$$

This reveals that the nuclear density ρ is approximately constant and independent of the mass number A. Therefore, the densities of copper $\left({}^{64}_{29}Cu\right)$ and carbon $\left({}^{12}_{6}C\right)$ nuclei are effectively the same because the density formula results in a similar value regardless of the specific mass number.

35. conceptual

36. .

$$a_c = \frac{v^2}{r}$$

$$F_G = \frac{GMm}{r^2}$$

$$u\sin g\ newton's\ \sec ond\ law$$

$$\frac{GMm}{r^2} = m\frac{v^2}{r}$$

$$v^2 = \frac{GM}{r} = \frac{gR^2}{r}$$

$$r^2 = \frac{9.8\times(6400)^2}{8000}$$

$$v = 7.08\ km/\sec$$

$$time\ period\ T = \frac{2\pi r}{V}$$

$$T = \frac{2\pi(8000km)}{7.08\ km/\sec}$$

$$= 118\ \min$$

37. ..

For More Join: @IITJEE_Advanced

step : 1 – *convert units and identify parameters*

first, *convert the plane's speed from kilometers per hour to meters per* second :

$v_x = 216km/h \times \frac{1000m}{3600s} = 60m/s$

given parameters are time $r = 10s$, *vertical displacement*

$y = 500m$, *and horizontal velocity* $v_x = 60m/s$.

Step – 2 : *Calculate horizontal displacement*

The package maintains a constant horizontal velocity

(*neglecting air resistance*), *so the horizontal displacement x is*

given by :

$x = v_x t$

$x = 60m/s \times 10s = 600m$

Step – 3 : *Calculate total displacement magnitude*

The total displacement 'P' from the release position has horizontal component x and vertical component x and vertical component y.The magnitude of the total displacement vector S is :

$S = \sqrt{x^2 + y^2}$

$S = \sqrt{(600m)^2 + (500m)^2} = \sqrt{360000 + 250000} = \sqrt{610000}$

$100\sqrt{61}$ *meters* (*approximately* 781.02 *meters*)

38..

If v is the velocity of CM of the body of radius r, then

Velocity at point A, $v_A = 0$

Velocity at point B, $v_B = v\sqrt{2}$

Velocity at point C, $v_C = v + r\omega = 2v$

Velocity at point D,

$V_D = \sqrt{V^2 + V^2 + 2V^2 \cos 120^0}$

$V_D = V$

Hencem

39. If a solid sphere placed in the fluid under high pressure, then it is compressed uniformly on all sides.

The force applied by the fluids acts in perpendicular direction at each point of the surface and the body is said to be under hydraulic compression.

This leads to decrease in its volume without any change of its geometrical shape.

Therefore, assertion and Reason are correct and Reason is the correct explanation of Assertion

40. Statement-II

During collision time, some K.E is stored as P.E in the form of deformation

41. .

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The ring is in contact with water along its inner and outer circumference. When pulled out, the total force on it due to surface tension will be

$$F = T(2\pi r_1 + 2\pi r_2) + mg$$

$$\Rightarrow \quad T = \frac{F - mg}{2\pi(r_1 + r_2)}$$

Given that an extra 3.97 gwt is required to pull it away from water so we have

$$F - mg = 3.97 \text{ gwt} = (3.97)(980) \text{ dyne}$$

$$\Rightarrow \quad T = \frac{3.97 \times 980}{3.14 \times (8.5 + 8.7)} = 72.13 \text{ dynecm}^{-1}$$

42. .

$$V = \sqrt{r\frac{P}{\rho}}$$

$$\Rightarrow v \alpha \sqrt{r}$$

$$\Rightarrow V_{Ne} : V_{Cl_2} : V_{CH_4} = \sqrt{V_{Ne}} : \sqrt{V_{Cl_2}} : \sqrt{V_{CH_4}}$$

$$= \sqrt{\frac{5}{3}} : \sqrt{\frac{7}{5}} : \sqrt{\frac{4}{3}}$$

43. .

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1. **Initial Moment of Inertia:**
   - For a uniform rod of mass M and length l about its perpendicular bisector, $I = \frac{1}{12} Ml^2$.

2. **Change in Length due to Temperature:**
   - When heated by ΔT, the length increases by $\Delta l = l\alpha\Delta T$, where α is the coefficient of linear expansion.

3. **New Length:**
   - The new length becomes $l' = l + \Delta l = l(1 + \alpha\Delta T)$.

4. **New Moment of Inertia (I'):**
   - $I' = \frac{1}{12} M(l')^2 = \frac{1}{12} M[l(1 + \alpha\Delta T)]^2 = \frac{1}{12} Ml^2(1 + 2\alpha\Delta T + (\alpha\Delta T)^2)$.

5. **Neglecting Higher-Order Terms:**
   - Since ΔT is small, $\alpha\Delta T$ is very small, and $(\alpha\Delta T)^2$ is negligible.
   - So, $I' \approx \frac{1}{12} Ml^2(1 + 2\alpha\Delta T) = \frac{1}{12} Ml^2 + \frac{1}{12} Ml^2(2\alpha\Delta T)$.

6. **Increase in Moment of Inertia (ΔI):**
   - $\Delta I = I' - I = \left(\frac{1}{12} Ml^2 + \frac{1}{12} Ml^2(2\alpha\Delta T)\right) - \frac{1}{12} Ml^2$.
   - $\Delta I = \frac{1}{12} Ml^2(2\alpha\Delta T) = \frac{1}{6} Ml^2\alpha\Delta T$.

7. **Expressing in terms of I:**
   - Since $I = \frac{1}{12} Ml^2$, then $2I = \frac{1}{6} Ml^2$.
   - Therefore, $\Delta I = (2I)\alpha\Delta T = 2I\alpha\Delta T$.

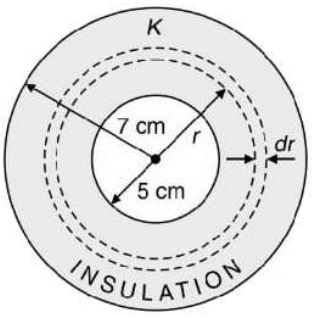
The increase in the moment of inertia is $\boldsymbol{\Delta I = 2I\alpha\Delta T}$ or $\boldsymbol{\frac{ML^2}{6}\alpha\Delta T}$.

44..

For More Join: @IITJEE_Advanced

Thermal resistance per meter length of an element at distance r of thickness dr is

$$dR = \frac{dr}{k(2\pi r)} \qquad \left\{\because R = \frac{l}{kA}\right\}$$



Total resistance $R = \int_{1=5\text{ cm}}^{2=7\text{ cm}} dR$

$$\Rightarrow \quad R = \frac{1}{2\pi k}\left[\int_{5\times10^{-2}}^{7\times10^{-2}} \frac{dr}{r}\right] = \frac{1}{2\pi k}\log_e\left(\frac{7}{5}\right)$$

$$\Rightarrow \quad R = \frac{1}{(2\pi)(0.07)}\log_e(1.4) = 0.765\ \text{KW}^{-1}$$

Since, Heat current $H = \dfrac{\text{Temperature difference}}{\text{Thermal resistance}}$

$$\Rightarrow \quad H = \frac{(100-20)}{0.765} = 104.6\ \text{watt}$$

So, heat lost Q in one hour is

$$Q = Ht = (104.6\ \text{Js}^{-1})(3600\ \text{s}) = 3.76\times10^5\ \text{J}$$

45. .

b)The strain produced by a hydraulic pressure is called volumetric strain as pressure created a normal force on every point and is defined as ratio of change in volume ΔV to the original volume V

i.e Volume strain= $\dfrac{\Delta V}{V}$

Therefore, Assertion and Reason are correct but Reason is not the correct explanation of Assertion

46. ..

For More Join: @IITJEE_Advanced

Let the momentum of e^- at any time t is p and its de-broglie wavelength is λ.

Then, $p = \frac{h}{\lambda}$

$$\frac{dp}{dt} = \frac{-h}{\lambda^2}\frac{d\lambda}{dt}$$

$$ma = F = -\frac{h}{\lambda}\frac{d\lambda}{dt} \quad [m = \text{mass of e}]$$

Where, -ve sign represents decrease in λ with time

$$ma = \frac{-h}{(h/p)^2}\frac{d\lambda}{dt}$$

$$a = -\frac{p^2}{mh}\frac{d\lambda}{dt}$$

Where, -ve sign represents decrease in λ with time

$$ma = \frac{-h}{(h/p)^2}\frac{d\lambda}{dt}$$

$$a = -\frac{p^2}{mh}\frac{d\lambda}{dt}$$

$$a = -\frac{mv^2}{h}\frac{d\lambda}{dt}$$

$$\frac{d\lambda}{dt} = -\frac{ah}{mv^2} \quad \text{.... (1)}$$

here, $a = \frac{qE}{m} = \frac{e}{m}\frac{\sigma}{2\varepsilon_0}$

$$a = \frac{\sigma e}{2\, m\varepsilon_0}$$

and $v = u + at$

$v = at$

Substituting values of a \& v in equation (1)

$$\frac{d\lambda}{dt} = -\frac{2\,h\varepsilon_0}{\sigma t^2}$$

$$\Rightarrow \frac{d\lambda}{dt} \propto \frac{1}{t^2}$$

$$\Rightarrow n = 2$$

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47. .

$V = \sqrt{\frac{B}{\rho}} \quad B = bulk\ modulus$

$given\ \ \rho_{gas} = 2\rho_{air}$

$assumed\ B = same$

$\frac{V_{gas}}{V_{air}} = \sqrt{\frac{\rho_{air}}{\rho_{gas}}} = \sqrt{\frac{\rho_{air}}{2\rho_{air}}} = \frac{1}{\sqrt{2}}$

$V_{gas} = \frac{V_{air}}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 212.3\,m/\sec$

$fundemental\ \ frequency\ \ f = \frac{V_{gas}}{2L}$

$L = length\ of\ \ pipe$

$f = \frac{300}{\sqrt{2}\times 2\times 1} = \frac{300}{2\sqrt{2}}$

$\sec ond\ \ harmonic\ \ f_2 = 2f_0 = 2\times\frac{300}{2\sqrt{2}} = \frac{300}{\sqrt{2}}$

$f_2 - f = \frac{300}{\sqrt{2}} - \frac{300}{2\sqrt{2}} = \frac{300}{\sqrt{2}}\left(1-\frac{1}{2}\right)$

$= \frac{300}{2\sqrt{2}} = \frac{150}{\sqrt{2}} = 106.08$

48. ..

$M = 12kg$

$m = 3kg$

$C.M\ \ of\ \ wheel\ when\ reached\ at\ bottom\ of\ \ inclined\ \ plane$

$PQ = \frac{1}{2}\sqrt{xgh} \text{ --------(1)}$

$For\ rolling\ wheel$

$\tau = I\alpha$

$(12g\sin\alpha - 3g\sin\alpha)R = \left(2\times 12R^2 + 3R^2\right)\frac{a}{R}$

$9g\ \sin\alpha\ \ R = 27R^2\times\frac{a}{R}$

$a = \frac{g\sin\alpha}{3}$

$V = \sqrt{2aS}$

$= \sqrt{2\frac{g\sin\alpha}{3}\times\frac{h}{\sin\alpha}}$

$= \sqrt{\frac{2gh}{3}}$

$= \frac{1}{2}\sqrt{\frac{8gh}{3}} \text{ ----------(2)}$

$comparing\ 1\ and\ \ 2$

$x = \frac{8}{3} = 2.67 \approx 3$

For More Join: @IITJEE_Advanced

49. .

$27 \times (4/3)\pi r^3 = (4/3)\pi R^3$

Or $r = R/3$

Terminal velocity, $v \propto r^3$

Therefore, $(v_1/v_2) = (R^2/r^2)$

$v_1/v_2 = [R/(R/3)]^2 = 9$

$(v_1/v_2) = 9$

50.

$\beta_0 = magnetic\ filed$

$speed\ of\ light\ C = 3\times10^8 m/\sec$

$\boxed{E_0 = CB_0}$ $B_0 = \frac{E_0}{C}$

$B_0 = \frac{120}{3\times10^8} = 4\times10^{-7} Tesla$

.

For More Join: @IITJEE_Advanced

CHEMISTRY

51. (d) $N_1V_1 = N_2V_2 + N_3V_3$

i.e. $0.1 \times 100 = 0.2 \times 30 + 0.25V_3$

$0.25V_3 = 4$

or $V_3 = 16$ mL

52. .

0 (d) The order of energy of orbitals can be calculated from $(n + l)$ rule. The lower the value of $(n + l)$ for an orbital, lower is its energy. If two orbitals have same $(n + l)$ value, the orbital with lower value of n has the lower energy.

(i) $6p = n + l = 6 + 1 = 7$

(ii) $5f = 5 + 3 = 8$

(iii) $4d = 4 + 2 = 6$

(iv) $5p = 5 + 1 = 6$

∴ The order of decreasing energy will be $5f > 6p > 5p > 4d$.

53. .

(a) Electronegativity values of given elements are as follows

Be-1.5(A), Mg-1.2(D), O-3.5(B), N-3.0(C)

Hence, the correct order

B > C > A > D.

54. .

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(*b*) NO contains 15 electrons.

Expanded octet molecule contains more than eight electrons in their valence shell. SF_6, PCl_5 and H_2SO_4 are examples of expanded octet molecules. BCl_3 and SF_6 are even electron molecules.

55. .

ΔH	ΔS	Comment on temperature (T)	ΔG	Comment on the process
< 0	> 0	at any temp.	< 0	spontaneous

56. .CONCEPTUAL

57. .

(*c*) Statement I is false but Statement II is true. The correct explanation is given as :

Addition of Cl_2 will take reaction in backward direction but equilibrium constant will not change as it depends only on temperature.

58. **CONCEPTUAL**

59. .

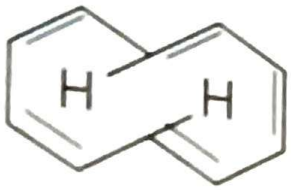
(*d*) Correct form is as follows

Among isomeric pentanes, 2, 2 - dimethylpentane has the lowest boiling point.

60. .

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5 (c) Among the given options only (c) is non-aromatic while other are aromatic.



Non - planar 10 πe^-
(Non - aromatic)

61. .

(c) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= -0.42 - (-0.72) = 0.3$

Net reaction,

$2Cr + 3Fe^{2+} \longrightarrow 2Cr^{3+} + 3Fe$

$\therefore \quad E = E^\circ_{cell} - \frac{0.0591}{6} \log \frac{[Cr^{3+}]^2}{[Fe^{2+}]^3}$

$= 0.26\ V$

62. .

(b) We know that,

$$r = k[A]^1[B]^1$$

From the given data

$$0.1 = k[20]^1[0.5]^1 \quad \ldots(i)$$

$$0.40 = k[x]^1[0.5]^1 \quad \ldots(ii)$$

$$0.80 = k[40]^1[y]^1 \quad \ldots(iii)$$

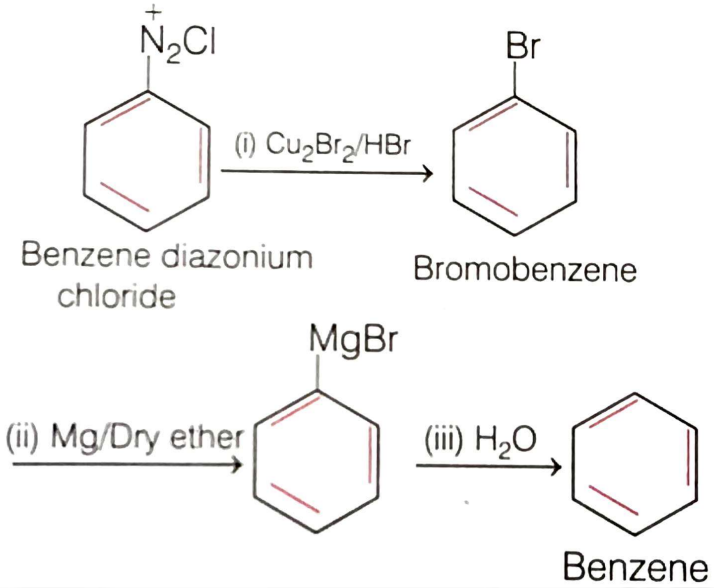
From Eqs. (i) and (ii) we will get $x = 80$ and from Eqs. (i) and (iii) we will get $y = 2$.

63. **CONCEPTUAL**

64. .

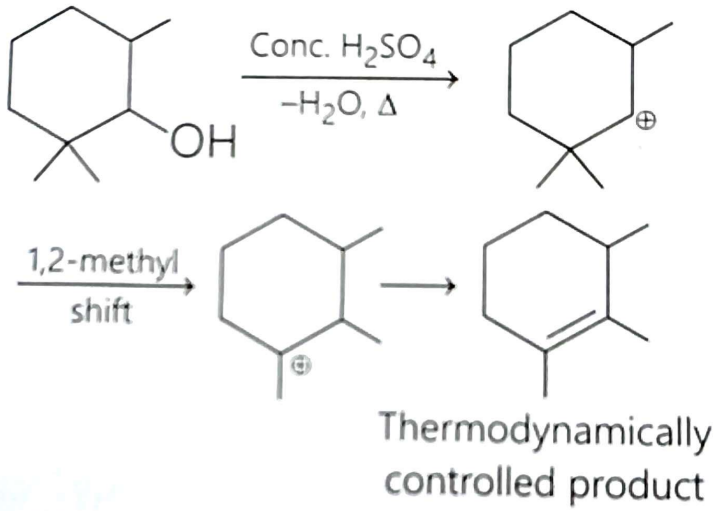
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(*a*) The reaction is proceed as



65. .

(*b*) The major product from the following reaction is given in option (b).



66. .

(*a*) Reducing agents like $LiAlH_4$, $NaBH_4$, i.e. complex hydrides usually does not affect π-bonds.

Thus, option (a) is correct answer.

67. .

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3 (b) The oxygen atom present in the ring shows *I*-effect.

As the distance between oxygen and —COOH group increases, –*I*-effect of oxygen decreases.

Thus, corresponding carboxylic acid will show less acidic nature.

The correct order of strengths of the carboxylic acids is (b).

68. .

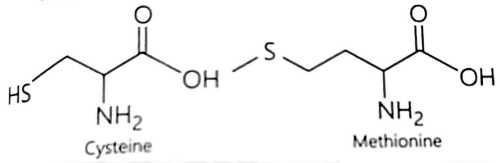
(d) The correct increasing order of basic strength is as follows

II < I < III.

Greater the electron density towards ring, greater will be basic strength of $—NH_2$.

69. .

16 (c) Sulphur containing amino acids are cysteine (II) and methionine (IV). Their structures are as follows



70. **CONCEPTUAL**

71. .

2 (c) Given,

(i) $C_2H_6(g) + \frac{1}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$,

$\Delta H^\circ_{comb} = -1560$ kJ/mol

(ii) $C(s) + O_2(g) \longrightarrow CO_2(g)$,

$\Delta H^\circ_{comb} = -394.0$ kJ/mol

(iii) $H_2(g) + \frac{1}{2}O_2 \longrightarrow H_2O(g)$,

$\Delta H^\circ_{comb} = -286.0$ kJ/mol

The reaction between carbon and hydrogen is :

$2C(s) + 3H_2(g) \longrightarrow C_2H_6(g)$,

$\Delta H^\circ_{r,n} = \Delta H^\circ_{f\,(C_2H_6,\,g)}$

$\Delta H^\circ_r = \Delta H^\circ_c$ (reactant) $-\Delta H^\circ_c$ (Product)

$= 2 \times (-394) + 3(-286) - 1560$

$= -788 - 858 + 1560$

$= -86.0$ kJ/mol

72. .

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3 (25) Less R_f value means more polar. Thus, among A, B and C, C will be most polar. Thus,

$$R_f = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$$

$$= \frac{2}{8} = 0.25 = 25 \times 10^{-2}$$

$$= \frac{10.2g}{176g\ mol^{-1}} = 0.048mol$$

$$\Delta T_f = K_f \times m$$

73. **Moles of ascorbic acid** $= 3.9 \times \frac{0.058}{150} \times 1000$

$$= 1.5^0 C = 15 \times 10^{-1}\ {}^0C$$

thus, the value of X is 15

74. **. For a first order reaction**

$$k = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$$

$$\frac{0.693}{10} = \frac{2.303}{100}\log\left(\frac{100}{100-x}\right)$$

$$x = 99.9\%$$

$$\approx 100\%$$

75. .

Complex	Valence shell configuration	Magnetic nature
$[Fe(CN)_6]^{4-}$	$3d^6$ (pairing)	Diamagnetic
$[Ti(CN)_6]^{3-}$	$3d^1$	Paramagnetic
$[Ni(CN)_4]^{2-}$	$3d^8$ (pairing)	Diamagnetic
$[Fe(CN)_6]^{3-}$	$3d^5$ (pairing)	Paramagnetic
$[Co(CN)_6]^{3-}$	$3d^6$ (pairing)	Diamagnetic

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